

Customer Churn Insights

Streamlit web dashboard for churn prediction, model comparison, explainability, and retention recommendations.

Customers

7,043

Churn rate

26.5%

Best model

Gradient Boosting

Best ROC-AUC

0.843

Prediction

Model Performance

Threshold Tuning

Business Insights

Customer Risk Prediction

Gender

Male

Phone service

Yes

Tech support

No

Senior citizen

No

Multiple lines

No

Streaming TV

No

Partner

Yes

Internet service

Fiber optic

Streaming movies

No

Dependents

No

Online security

No

Paperless billing

Yes

Tenure in months

3

Online backup

No

Payment method

Electronic check

Contract

Two year

Device protection

No

Monthly charges

35.00

Total charges

105.00

Decision threshold

0.50

Predict churn risk

Churn probability

14.8%

Risk level

Low

Customer currently shows limited churn risk.

Predicted class: no churn

Decision threshold: 0.50

Input customer profile																				
	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	
0	Male	0	Yes	No	3	Yes	No	Fiber optic	No	No	No	No	No	No	Two year	Yes	Electronic check	35	105	